

Md. Almabud **JUWAL**
Senior Software Engineer | B.Sc. Computer Science and Engineering

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 Dhaka, Bangladesh

 EDUCATION

2020	B.Sc. in Computer Science and Engineering
2016	> Daffodil International University, Dhaka, Bangladesh > CGPA : 3.77 / 4.00 > Relevant coursework : Artificial Intelligence, Algorithms, Data Structures, Software Engineering, Compiler Design, Discrete Mathematics, Computer Networks

 ACADEMIC & RESEARCH INTERESTS

INTERESTS : Artificial Intelligence, Machine Learning, Data Science, Software Engineering, Distributed Systems

 SKILLS

Programming	Python, SQL, JavaScript/TypeScript, Go, C/C++
Backend & Frameworks	FastAPI, Django, Django REST Framework, Celery
Data & Databases	PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch
Infrastructure & Engineering	Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible
Frontend	Angular

 EXPERIENCE

Present Jun 2024	Senior Software Engineer Cogwheel Analytics, Colorado, United States > Architected and led the development of the next-generation Cogwheel Analytics platform as a modular monolith following Clean Architecture principles, with clear separation of concerns and reusable handling for authentication, pagination, and request validation, reducing repetitive implementation work and accelerating feature delivery > Lead the development and maintenance of the Data Cleaning service, implementing data-validation and transformation workflows to improve data integrity and consistency > Design and implement backend services and APIs using Python, FastAPI, and PostgreSQL, with emphasis on maintainability, reliability, and performance > Build and maintain asynchronous data-processing and web-scraping workflows using Pandas, Celery, and Scrapy
May 2024 Jul 2023	Senior Software Engineer Brainstation 23 Ltd, Dhaka, Bangladesh > Led the turnaround of a delayed project by mentoring 3 junior engineers, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within 6 months > Designed software architectures for Task Station and Sayvery, with focus on scalability, maintainability, and clear separation of responsibilities > Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines > Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction

Jun 2023 Oct 2021	Software Engineer Brainstation 23 Ltd, Dhaka, Bangladesh ➤ Improved system performance by approximately 10% through database-query optimization ➤ Improved API response performance by approximately 5% by introducing application-level caching ➤ Upgraded Python and Django versions and refactored the existing codebase to improve maintainability and compatibility ➤ Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer
Sep 2021 Jul 2020	Junior Software Engineer Tirezok Private Limited, Dhaka, Bangladesh ➤ Reduced processing time for 3.3 million records from approximately 29–35 minutes to 2–4 minutes using PostgreSQL COPY and database triggers ➤ Reduced database query response time by approximately 50% through query optimization ➤ Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j ➤ Automated deployment workflows using Jenkins and Ansible, reducing manual deployment effort and improving release consistency
Apr 2020 Dec 2019	Web Developer Wizdoor Inc, Dhaka, Bangladesh ➤ Contributed to backend and database architecture while coordinating implementation with frontend and design teams ➤ Developed a real-time chat application using Django Channels and WebSockets

PROJECTS

COGWHEEL ANALYTICS

Jun 2024 – Present

<https://cogwheelanalytics.com>

Enterprise analytics platform for automated data ingestion, validation, processing, and reporting. The next-generation platform is designed as a modular monolith following Clean Architecture principles, with reusable application components and asynchronous processing to support maintainable and scalable analytics workflows.

REHIVE

Oct 2021 – Feb 2023

<https://rehive.com/>

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

LEGISLIB

May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

ACADEMIC & PROGRAMMING ACHIEVEMENTS

2018	Participant, ACM ICPC Dhaka Preliminary Contest
2018	4th place , DIU Intra Team Programming Contest
2017	2nd place , DIU Intra Team Programming Contest
2017	9th place , DIU Takeoff Individual Programming Contest

CERTIFICATIONS & COURSES

Jun 2025	Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka
Apr 2023	AWS Certified Cloud Practitioner
2021	Object-Oriented Design, Coursera
2021	The Ultimate Docker Course, Mosh Hamedani

TEACHING AND MENTORING

Dec 2018	ACM Co-Chair Computer & Programming Club, Daffodil International University
Jan 2018	
	> Coordinated programming contests and other ACM-related club activities > Mentored students in competitive programming and encouraged participation in programming contests
Jan 2019	Student Prefect Daffodil International University
May 2017	
	> Assisted students with Competitive Programming and Algorithms through practical lab sessions and academic support > Assisted the course instructor with laboratory examinations and student performance evaluation > Mentored students requiring additional academic guidance



LANGUAGES

- > **Bengali** — Native
- > **English** — B2 (CEFR), IELTS Academic 6.5
- > **Hindi** — Conversational